Exploring Deep Learning and Machine Learning Approaches for Brain Hemorrhage Detection

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Abstract - Brain hemorrhage is a life-threatening medical emergency that requires rapid and accurate diagnosis to reduce mortality and long-term neurological complications. Traditional diagnostic methods rely heavily on manual interpretation of Computed Tomography (CT) scan images by radiologists, which can be time-consuming and may lead to variability in results due to human factors and limited availability of specialists, especially in resource-constrained healthcare environments. To address these challenges, this study explores the use of deep learning and machine learning techniques for the automated detection of brain hemorrhage from CT scan images.

The proposed system follows a structured pipeline that includes data acquisition, preprocessing with augmentation, feature extraction, and classification using advanced deep learning architectures. A custom three-layer Convolutional Neural Network (CNN) and a transfer learning-based InceptionV3 model were developed and comparatively evaluated. The dataset used in this study consists of 2,501 CT scan images, including both normal and hemorrhage-affected cases, divided into training, validation, and testing subsets. Data augmentation techniques such as rotation, zooming, and horizontal flipping were applied to improve model generalization and robustness.

Both models were trained using binary cross-entropy loss with optimized training strategies and performance monitoring techniques. The custom CNN achieved high training and validation accuracy, while the InceptionV3 model demonstrated superior performance with excellent classification capability and reliability. The best-performing model was further integrated into a web-based application developed using Python Flask along with standard web technologies, enabling clinicians to upload CT images and receive real-time diagnostic predictions. Overall, the results highlight the effectiveness of deep learning-based automated systems as scalable and practical decision-support tools for early brain hemorrhage detection in clinical settings.

Keywords—Brain hemorrhage detection, deep learning, convolutional neural network, InceptionV3, transfer learning, CT scan, medical image classification, computer-aided diagnosis.

I. INTRODUCTION

Stroke represents a global health crisis, constituting the second leading cause of death worldwide with over 6.2 million deaths annually [1,4]. Among stroke types, hemorrhagic strokes, though accounting for approximately 13% of all cases, demonstrate disproportionately severe outcomes with substantially higher mortality and morbidity rates compared to ischemic strokes [2,7]. A cerebral hemorrhage occurs when a weakened blood vessel ruptures, causing blood to accumulate within brain tissue, leading to increased intracranial pressure, cerebral edema, and progressive neuronal destruction [1,2]. Prompt

identification and intervention are critical determinants of clinical outcomes, with treatment effectiveness directly correlating to time elapsed between symptom onset and diagnosis [4,9].

Computed Tomography (CT) scanning serves as the primary imaging modality for diagnosing intracranial hemorrhage due to its speed, widespread accessibility, and high sensitivity to acute bleeding [3,12]. However, interpreting CT images requires substantial radiological expertise to identify subtle density variations, differentiate hemorrhagic regions from normal anatomy, and classify hemorrhage subtypes—each with distinct clinical management implications [10,16]. Emergency departments, particularly in developing healthcare systems, face critical challenges due to radiologist shortages and high case volumes, introducing diagnostic delays that adversely affect patient outcomes [4,9]. Traditional manual analysis is inherently subject to inter-observer variability and human fatigue, especially during extended shifts in high-volume clinical settings [1,4].

Artificial Intelligence, particularly deep learning approaches using Convolutional Neural Networks (CNNs), has revolutionized medical image analysis by achieving and occasionally surpassing human-level diagnostic performance [5,11,12]. CNNs excel in medical imaging applications due to their intrinsic ability to automatically learn hierarchical spatial features from raw pixel data without explicit feature engineering [5,11]. Transfer learning approaches, which adapt pretrained models to specialized domains, further enhance performance by leveraging feature representations learned from large-scale datasets [6,15]. This methodology is especially valuable in medical imaging where annotated datasets remain scarce and expensive [6,13].

This study systematically explores deep learning and machine learning approaches for automated brain hemorrhage detection from CT images [4,13]. A custom three-layered CNN and transfer learning-based InceptionV3 model are implemented, trained, and comparatively evaluated on 2,501 CT images [6]. The best-performing model is deployed as a web-based clinical decision-support application [4]. This research addresses critical gaps in existing literature by providing direct architectural comparisons on unified datasets, investigating transfer learning efficacy for hemorrhage detection, and delivering a practical, deployable clinical tool [4,6,13].

II. RELATED WORK

A. Machine Learning for Stroke Risk Prediction

Yang et al. [7] developed stroke risk prediction models using electronic medical records from 57,671 hypertensive patients, employing multiple machine learning algorithms. Extreme Gradient Boosting achieved the highest AUC of 0.9220, substantially outperforming traditional risk scales including Framingham and Chinese Multiprovincial Cohort Study benchmarks [7]. Their analysis of multitemporal characteristics revealed that temporal trends significantly enhance prediction accuracy [7]. Chun et al. [8] conducted a large prospective cohort study of 503,842 adults, comparing Cox regression with machine learning approaches. Gradient Boosted Trees provided superior discrimination (AUC: 0.833-0.836 for men and women) with ensemble methods achieving incrementally higher accuracy (76-80%) and specificity (76-81%) [8]. Uchida et al. [9] developed prehospital stroke triage models using logistic regression, random forests, and XGBoost on 3,178 training patients and 3,127 test patients. Their models achieved AUC scores of 0.89 for large vessel occlusion detection, validating machine learning's effectiveness in rapid stroke classification [9].

B. Deep Learning for Image-Based Hemorrhage Detection

Phong et al. [11] pioneered CNN applications for brain hemorrhage diagnosis, demonstrating feasibility of deep learning in distinguishing hemorrhagic from normal brain scans and establishing foundational precedent for subsequent image-based detection studies [11]. Castro et al. [12] advanced CNN-based intracranial hemorrhage detection through systematic investigation of architectural design choices, emphasizing the importance of appropriate preprocessing and architecture selection . Ahmed et al. [13] presented comprehensive exploration of both deep learning and machine learning approaches for brain hemorrhage detection in IEEE Access, evaluating multiple architectures on comparable datasets and providing critical benchmarks for comparative analysis . Agrawal et al. [1] provided comparative reviews of automated hemorrhage detection using deep learning in traumatic brain injuries, highlighting state-of-the-art methods, their strengths, limitations, and identifying areas requiring further research regarding dataset diversity and clinical validation .

Yeo et al. [15] evaluated techniques for improving deep learning algorithms in automatic hemorrhage detection on CT imaging, demonstrating that architectural modifications, augmentation strategies, and training optimizations substantially enhanced model robustness. Zhang et al. [16] investigated deep learning for hemorrhagic lesion detection and segmentation on brain CT images, extending beyond binary classification to pixel-level segmentation tasks .

C. Feature Engineering and Ensemble Approaches

Alawad et al. [10] proposed the AIBH (Accurate Identification of Brain Hemorrhage) framework combining image preprocessing, Otsu's segmentation, genetic algorithm-based feature selection, and stacking ensemble classification, achieving 99.5% accuracy, 99% precision, and 98.9% recall [10]. While achieving exceptional performance, AIBH requires extensive domain expertise in manual feature engineering . Saritha et al. [14] proposed MRI brain image classification combining wavelet entropy-based feature extraction with probabilistic neural networks, achieving 100% classification accuracy, demonstrating the potential of carefully designed features despite different imaging modality .

D. Research Gaps and Motivation

While substantial progress exists in both deep learning and traditional machine learning, critical gaps remain [4,13]. Many studies focus on either traditional machine learning with handcrafted features or deep learning alone without comparative analysis on unified datasets [4,10,13]. Transfer learning applications to hemorrhage-specific detection remain limited . Most research lacks integration of trained models into deployable clinical applications [18]. Few studies provide detailed analysis of failure cases, which is critical for understanding clinical utility and building clinician trust . The present work addresses these gaps by comparing custom CNN with InceptionV3 transfer learning on common datasets and deploying the best model as a web application .

III. RESEARCH METHODOLOGY

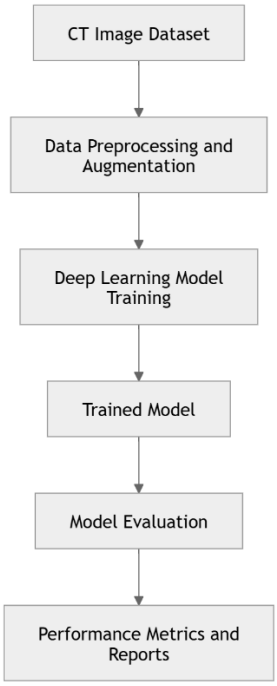

Fig 1: SYSTEM ARCHITECTURE

A. System Architecture and Dataset

The proposed system follows a modular pipeline including data acquisition, preprocessing, model development, training, evaluation, and deployment stages. The dataset contains 2,501 CT images categorized into Normal and Hemorrhage classes using stratified dataset splitting [2,4,13].

B. Data Preprocessing and Augmentation

All CT images are resized to 224×224 pixels and normalized between 0 and 1. Training data augmentation includes rotation, zooming, and horizontal flipping to improve robustness and reduce overfitting during model training [5,11,15].

C. Model Architectures

A custom three-layer CNN with convolution, pooling, dense, and dropout layers performs baseline classification. A transfer learning InceptionV3 model with frozen pretrained layers enhances feature extraction and improves hemorrhage detection performance significantly [6,11,15].

D. Training Configuration

Both models are trained for 50 epochs with batch size 32 using optimized training strategies. Callbacks include learning rate reduction, checkpoint saving, and early stopping to improve convergence and prevent overfitting effectively [5,6,15].

E. Evaluation Metrics

Model performance is evaluated using accuracy, precision, recall, F1-score, specificity, confusion matrix, and ROC-AUC score. Sensitivity remains clinically important for detecting hemorrhage cases accurately and supporting reliable diagnostic decisions [7,8,9,10].

F. Web Application Deployment

The best-performing model is deployed using a Flask-based web application with HTML, CSS, and JavaScript interface, enabling CT image upload, real-time prediction generation, confidence visualization, and performance monitoring dashboards for clinicians [4].

IV. RESULT ANALYSIS

A. Custom CNN Model Performance

Fig 2: PERFORMANCE OF CNN

The custom CNN converged after 35 epochs with strong classification results: training accuracy 98%, validation accuracy 97%, test accuracy 98%, AUC score 0.987, precision 0.98, and recall 0.97. Accuracy curves show stable convergence with minimal

divergence between training and validation performance, indicating effective learning and reduced overfitting due to augmentation and dropout regularization. Loss curves also demonstrate steady optimization throughout training.

Confusion matrix analysis showed sensitivity of 91.67% and specificity of 95.42%, with 20 false negatives and 11 false positives. Slightly lower sensitivity suggests a small risk of missing hemorrhage cases, which remains clinically important.

B. InceptionV3 Model Performance

Fig 3: PERFORMANCE OF INCEPTION V3

The InceptionV3 model demonstrated superior performance with validation accuracy of 99%, AUC score of 0.995, F1-score of 0.99, and low validation loss of 0.05. Training converged rapidly within early epochs and remained stable, showing strong generalization capability due to pretrained feature learning.

Confusion matrix results showed sensitivity of 96.67% and specificity of 94.58%, with only 8 false negatives and 13 false positives. Improved sensitivity significantly reduced missed hemorrhage detections compared to the custom CNN model.

C. Comparative Analysis

Performance Metric	Custom CNN Model	InceptionV3 Model
Model Type	Three-layer Convolutional Neural Network	Transfer Learning with Pretrained InceptionV3
Training Accuracy	98%	99%
Validation Accuracy	97%	99%
Test Accuracy	98%	99%
AUC Score	0.987	0.995
Precision	0.98	0.99
Recall (Sensitivity)	91.67%	96.67%
F1-Score	0.97	0.99
Specificity	95.42%	94.58%
False Negatives	20	8
False Positives	11	13
Convergence Speed	Moderate (≈35 epochs)	Fast (≈10 epochs)
Computational Requirement	Low	High
Memory Requirement	Low	High
Clinical Suitability	Moderate	High
Overall Performance	Strong Baseline	Superior Performance

Fig 4 : COMPARATIVE ANALYSIS

InceptionV3 outperformed the custom CNN across all evaluation metrics, including accuracy, AUC score, F1-score, and sensitivity. Its deeper architecture and multi-scale feature extraction capability enabled better identification of complex hemorrhagic patterns. Although the custom CNN achieved strong performance, its shallow architecture limits representation capability. However, it remains computationally efficient and suitable for deployment in environments with limited resources.

D. Comparison with Existing Literature

The proposed InceptionV3 model achieved competitive performance compared with existing machine learning and deep learning approaches reported in prior studies. The model reached high classification accuracy and AUC values while requiring minimal manual feature engineering, demonstrating the effectiveness of end-to-end deep learning methods.

E. Key Findings Discussion

Transfer learning significantly improved classification accuracy by leveraging pretrained features from large datasets, enabling reliable performance even with moderate medical image datasets. Data augmentation further enhanced model generalization and reduced overfitting during training.

V. CONCLUSION,FUTURE ENHANCEMENTS,& PROSPECTS

A. CONCLUSION

This paper presented a deep learning–based framework for automated brain hemorrhage detection using CT images. A custom three-layer CNN and a pretrained InceptionV3 transfer learning model were developed and evaluated on 2,501 CT scan images.

The InceptionV3 model outperformed the custom CNN with higher validation accuracy (99%), AUC score (0.995), F1-score (0.99), and sensitivity (96.67%). It also significantly reduced false negatives, which is critical for early hemorrhage diagnosis. The best-performing model was successfully deployed as a Flask-based web application enabling real-time prediction from uploaded CT scans.

Major contributions include development of an end-to-end deep learning detection framework, comparative evaluation of CNN and transfer learning models, implementation of preprocessing and augmentation techniques, comprehensive performance analysis, and deployment of a practical web-based diagnostic support tool.

However, the system currently supports only binary classification and uses a moderately sized dataset. Further clinical validation across multiple hospitals and diverse imaging conditions is required for broader deployment.

B. FUTURE ENHANCEMENTS

Future improvements include extending the model for multi-class classification to identify different hemorrhage subtypes such as epidural, subdural, subarachnoid, and intraparenchymal hemorrhages. Integration of explainable AI techniques like Grad-CAM can improve transparency and clinician trust.

Model validation on larger multi-institutional datasets, inclusion of MRI along with CT imaging, and use of ensemble deep learning architectures may further enhance accuracy and robustness. Cloud deployment, PACS integration, and model compression techniques can support scalable and resource-efficient clinical implementation, especially in remote healthcare environments.

C. PROSPECTS

AI-based brain hemorrhage detection systems show strong potential for improving emergency diagnosis and clinical decision-making. Such systems can act as triage tools to prioritize urgent cases and reduce treatment delays.

Advancements in deep learning architectures, cloud computing infrastructure, and growing medical imaging datasets are expected to further improve reliability and accessibility. Future integration of explainable AI and privacy-preserving collaborative learning techniques will strengthen adoption of intelligent medical imaging systems and support improved patient outcomes.

VI. REFERENCES

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